MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – A)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

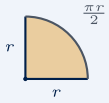
Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE QUESTIONS MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The perimeter of a quadrant of a circle of radius r is: [1]

(A) $\frac{\pi r}{2}$ (B) $2\pi r$ (C) $\frac{\pi r}{2} + 2r$ (D) $\pi r + 2r$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Arc Length of Quadrant [0.5 Mark]

A quadrant has central angle $\theta = 90°$.

$$\text{Arc length } l = \frac{90°}{360°} \times 2\pi r = \frac{1}{4} \times 2\pi r = \frac{\pi r}{2}$$

Step 2: Total Perimeter [0.5 Mark]

The perimeter consists of the arc length plus two bounding radii:

$$\text{Perimeter} = l + 2r = \frac{\pi r}{2} + 2r$$

Correct Option: (C) $\frac{\pi r}{2} + 2r$

$$\boxed{\text{Perimeter of quadrant} = \frac{\pi \mathbf{r}}{\mathbf{2}} + \mathbf{2r}}$$

Q2. What is the area of a semi-circle of diameter d? [1]

(A) $\frac{1}{16}\pi d^2$ (B) $\frac{1}{4}\pi d^2$ (C) $\frac{1}{8}\pi d^2$ (D) $\frac{1}{2}\pi d^2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Radius in terms of Diameter [0.5 Mark]

$$\text{Radius } r = \frac{d}{2}$$

Step 2: Area of Semicircle [0.5 Mark]

$$\text{Area} = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi\left(\frac{d}{2}\right)^2 = \frac{1}{2}\pi\left(\frac{d^2}{4}\right) = \frac{1}{8}\pi d^2$$

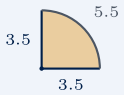
Correct Option: (C) $\frac{1}{8}\pi d^2$

$$\boxed{\text{Area of semicircle} = \frac{\mathbf{1}}{\mathbf{8}}\pi\mathbf{d^2}}$$

Q3. The perimeter of a quadrant of a circle of circumference 22 cm is: [1]

(A) 29 cm (B) 22 cm (C) 12.5 cm (D) 15.5 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Radius from Circumference [0.5 Mark]

$$C = 2\pi r = 22 \implies 2 \times \frac{22}{7} \times r = 22 \implies r = \frac{7}{2} = 3.5 \text{ cm}$$

Step 2: Perimeter of Quadrant [0.5 Mark]

$$\text{Arc length } l = \frac{C}{4} = \frac{22}{4} = 5.5 \text{ cm}$$

$$\text{Perimeter} = l + 2r = 5.5 + 2(3.5) = 5.5 + 7 = 12.5 \text{ cm}$$

Correct Option: (C) 12.5 cm

$$\boxed{\text{Perimeter of quadrant} = \mathbf{12.5}\text{ cm}}$$

Q4. The area of a sector of a circle of radius 16 cm cut off by an arc of length 18.5 cm is: [1]

(A) 168 cm^2 (B) 148 cm^2 (C) 154 cm^2 (D) 176 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Direct Sector Formula relating Arc Length and Radius [0.5 Mark]

$$\text{Area of sector} = \frac{1}{2}lr$$

Step 2: Substitution [0.5 Mark]

$$\text{Area} = \frac{1}{2} \times 18.5 \times 16 = 18.5 \times 8 = 148 \text{ cm}^2$$

Correct Option: (B) 148 cm^2

$$\boxed{\text{Area of sector} = \mathbf{148}\text{ cm}^\mathbf{2}}$$

Q5. **Assertion (A):** If the circumference of a circle is 176 cm, then its radius is 28 cm.
Reason (R): Circumference of a circle $= 2\pi r$, where r is the radius of the circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verification of Assertion (A) [0.5 Mark]

$$C = 2\pi r = 176 \implies 2 \times \frac{22}{7} \times r = 176 \implies \frac{44}{7} r = 176 \implies r = \frac{176 \times 7}{44} = 28 \text{ cm}$$

Hence, Assertion (A) is **True**.

Step 2: Verification of Reason (R) [0.5 Mark]

Reason (R) states the valid mathematical formula $C = 2\pi r$ which directly explains Assertion (A). Hence, (R) is **True** and correctly explains (A).

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Option (A) is correct

SECTION B: Short Answer Questions

MATHS MANTHAN | **(3 × 2 = 6 Marks)**

Q6. The minute hand of a clock is 14 cm long. Find the area on the face of the clock described by the minute hand in 5 minutes. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Central Angle for 5 Minutes [1 Mark]

In 60 minutes, the minute hand completes 360°.

$$\text{Angle swept in 1 minute} = \frac{360^\circ}{60} = 6^\circ$$

$$\text{Angle swept in 5 minutes } \theta = 5 \times 6^\circ = 30^\circ$$

Step 2: Area Described [1 Mark]

Length of minute hand $r = 14$ cm.

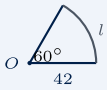
$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{30^\circ}{360^\circ} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{12} \times 22 \times 28$$

$$= \frac{1}{12} \times 616 = \frac{154}{3} \approx 51.33 \text{ cm}^2$$

$$\boxed{\text{Area described in 5 minutes} = \frac{154}{3} \text{ cm}^2 \approx \mathbf{51.33} \text{ cm}^2}$$

Q7. Find the length of the arc of a circle which subtends an angle of 60° at the centre of the circle of radius 42 cm. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[2]**

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Arc Length Formula [1 Mark]

Given radius $r = 42$ cm, central angle $\theta = 60°$.

$$l = \frac{\theta}{360°} \times 2\pi r$$

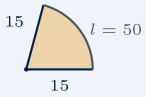
Step 2: Substitution and Calculation [1 Mark]

$$l = \frac{60°}{360°} \times 2 \times \frac{22}{7} \times 42 = \frac{1}{6} \times 2 \times 22 \times 6 = 44 \text{ cm}$$

Length of the arc = **44** cm

Q8. The perimeter of a sector of a circle of radius 15 cm is 80 cm. Find the area of the sector. **[2]**

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Finding Arc Length l [1 Mark]

Radius $r = 15$ cm. Perimeter of sector $= l + 2r = 80$ cm.

$$l + 2(15) = 80 \implies l + 30 = 80 \implies l = 50 \text{ cm}$$

Step 2: Area of Sector [1 Mark]

$$\text{Area of sector} = \frac{1}{2}lr = \frac{1}{2} \times 50 \times 15 = 25 \times 15 = 375 \text{ cm}^2$$

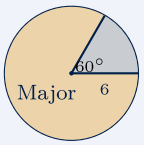
Area of the sector = **375** cm^2

SECTION C: Long Answer Questions

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q9. Find the area of the minor and the major sectors of a circle with radius 6 cm, if the angle subtended by the minor arc at the centre is 60°. [Use $\pi = 3.14$] **[3]**

► **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Area of Minor Sector [1.5 Marks]
Radius $r = 6$ cm, central angle $\theta = 60°$.

$$\text{Area}_{\text{minor}} = \frac{60°}{360°} \times \pi r^2 = \frac{1}{6} \times 3.14 \times 6^2 = \frac{1}{6} \times 3.14 \times 36 = 3.14 \times 6 = 18.84 \text{ cm}^2$$

Step 2: Area of Major Sector [1.5 Marks]

$$\text{Total Area of circle} = \pi r^2 = 3.14 \times 36 = 113.04 \text{ cm}^2$$

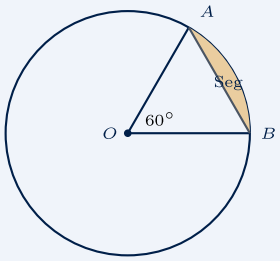
$$\text{Area}_{\text{major}} = \text{Total Area} - \text{Area}_{\text{minor}} = 113.04 - 18.84 = 94.2 \text{ cm}^2$$

(Alternatively: $Area_{major} = \frac{360° - 60°}{360°} \times \pi r^2 = \frac{300°}{360°} \times 113.04 = \frac{5}{6} \times 113.04 = 94.2\ cm^2$*.)*

Minor Sector Area = **18.84** cm^2, Major Sector Area = **94.2** cm^2

Q10. A chord of a circle of radius 10 cm subtends an angle of 60° at the centre of the circle. Find the area of the corresponding minor segment of the circle. [Use $\pi = 3.14$ and $\sqrt{3} = 1.73$] **[3]**

► **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Area of Sector OAB [1 Mark]
Radius $r = 10$ cm, central angle $\theta = 60°$.

$$\text{Area}_{\text{sector}} = \frac{60°}{360°} \times \pi r^2 = \frac{1}{6} \times 3.14 \times 10^2 = \frac{314}{6} = 52.33 \text{ cm}^2$$

Step 2: Area of $\triangle OAB$ [1 Mark]
In $\triangle OAB$, $OA = OB = 10$ cm and $\angle AOB = 60° \implies \triangle OAB$ is equilateral.

$$\text{Area}(\triangle OAB) = \frac{\sqrt{3}}{4} r^2 = \frac{1.73}{4} \times 100 = 1.73 \times 25 = 43.25 \text{ cm}^2$$

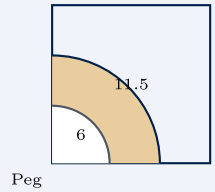
Step 3: Area of Minor Segment [1 Mark]

$$\text{Area}_{\text{segment}} = \text{Area}_{\text{sector}} - \text{Area}(\triangle OAB) = 52.33 - 43.25 = 9.08 \text{ cm}^2$$

Area of the minor segment = **9.08** cm^2

Q11. A horse is tied with a rope of length 6 m at the corner of a square grassy lawn of side 20 m. If the length of the rope is increased by 5.5 m (so the new length of rope is 11.5 m), find the increase in the area of the lawn in which the horse can graze. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Identifying Grazing Geometries [1 Mark]

At the corner of a square lawn, the corner angle is $90°$ (quadrant).

Initial radius $r_1 = 6$ m, New radius $r_2 = 6 + 5.5 = 11.5$ m

Step 2: Increase in Grazing Area Formula [1 Mark]

$$\Delta\text{Area} = \frac{1}{4}\pi(r_2^2 - r_1^2) = \frac{1}{4} \times \frac{22}{7} \times (11.5^2 - 6^2)$$

Step 3: Algebraic Evaluation [1 Mark]

$$\Delta\text{Area} = \frac{1}{4} \times \frac{22}{7} \times (11.5 - 6)(11.5 + 6) = \frac{1}{4} \times \frac{22}{7} \times 5.5 \times 17.5$$

$$= \frac{1}{4} \times 22 \times 5.5 \times 2.5 = \frac{11}{2} \times 13.75 = \frac{151.25}{2} = 75.625 \text{ m}^2$$

Increase in grazing area = **75.625 m²**

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★